Daily Practice Problem (DPP)

Graph Theory

Name: ______________________________

Q1. Draw these graphs.

(a) K_7 (b) $K_{1,8}$ (c) $K_{4,4}$ (d) C_7 (e) W_7

Q2. How many vertices and how many edges do these graphs have?

(a) K_n (b) C_n (c) W_n (d) $K_{m,n}$ (e) Q_n

Q3. For which values of n are these graphs bipartite?

(a) K_n (b) C_n (c) W_n (d) Q_n

Q4. The *degree sequence* of a graph is the sequence of the degrees of its vertices listed in nonincreasing order. Now, answer the following questions:

(a) Find the degree sequences of the following graphs:

(a) K_4 (b) C_4 (c) W_4 (d) $K_{2,3}$ (e) Q_3 (f) $K_{m,n}$ (g) K_n

(b) How many edges does a graph have if its degree sequence is. Also draw their corresponding graph.

(a) $4, 3, 3, 2, 2$ (b) $5, 2, 2, 2, 2, 1$

Q5. Let G be a graph with v vertices and e edges. Let M be the maximum degree of the vertices of G, and let m be the minimum degree of the vertices of G. Show that

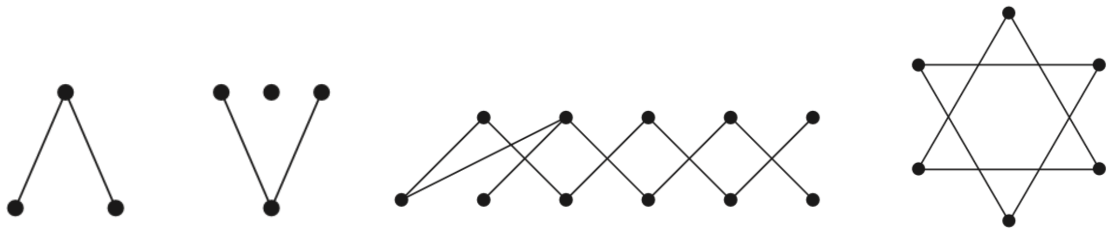
(a) $\dfrac{2e}{v} \geq m$ (b) $\dfrac{2e}{v} \leq M.$

Q6. A simple graph is called *regular* if every vertex has the same degree. A regular graph is called n-*regular* if every vertex has degree n.

- Determine the values of n for which the graphs K_n, C_n, W_n, and Q_n are regular.
- Determine for which values of m and n the bipartite graph $K_{m,n}$ is regular.
- Find the number of vertices in a regular graph of degree 4 having 10 edges.

Q7. Show that if G is a bipartite simple graph with v vertices and e edges, then $e \leq \frac{v^2}{4}$.

Q8. In each of the following questions, determine whether the given graph is connected or not. How many connected components does each of the graphs have? For each graph find each of its connected components.



(a) (b) (c)